

The Future is Red: Visualizing Wildfire Predictions using Contactless Interaction

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Figure 1: *The Future is Red* on the view at an exhibition in ARKO Art Center, Seoul, S. Korea in 2021

ABSTRACT

This paper presents our approach for visualizing, sonifying, and predicting wildfire in the near future and contactless interaction. We provide an interaction tool to depict the causes and results of the wildfire and promote awareness of the environmental issues by giving forecasting results of the wildfire to the audience. Multi-modal interaction allows the audience to dynamically experience the changes of the wildfire over time in two representative locations. (California, United States, and South Korea) The interactive multi-modal data visualization and sonification depict the past, present, and future of the wildfire. This data-driven design was installed in an art gallery and presented to audience members. Contactless user interaction with Leap Motion cameras was used during the pandemic period for hygienic interaction. In this paper, we describe the design process and how this interface was developed based on environmental issues, and informal user responses from the art gallery setup are discussed.

CCS CONCEPTS

• **Human-centered computing**; • **Visualization**; • **Visualization application domains**; • **Information visualization**;

KEYWORDS

wildfire, data visualization, data sonification, machine learning, predictive analysis, contactless interaction

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1 INTRODUCTION

California is currently marking a new record in the history of wildfire with the most critical impact on nature and the environment. [1] Extreme drought conditions, high winds, and dry air have impacted the animals, plants, mountainous topography, residential area, and even human population growth. The reduced moisture content of the severe drought area has circled back and impacted the wildfires. Many researchers and scientists have been investigating these issues and trying to find solutions. We wanted to create a data visualization to depict the current situations of the wildfire and correlated data to promote awareness of drought, wildfire, and air quality issues. On top of those approaches, more intuitive and engaging data visualization to visualize the past, present, and even future of wildfire was the main goal of this project. Furthermore, visualizing data is an effective way to understand the narrative, data sonification can support the perception and understanding of the data. Multi-sensory data representations and fabrication techniques have been used in data visualization these days like several physical data sculptures. [9] [10] New audiovisual aesthetics of data representations driven from the various materials and techniques

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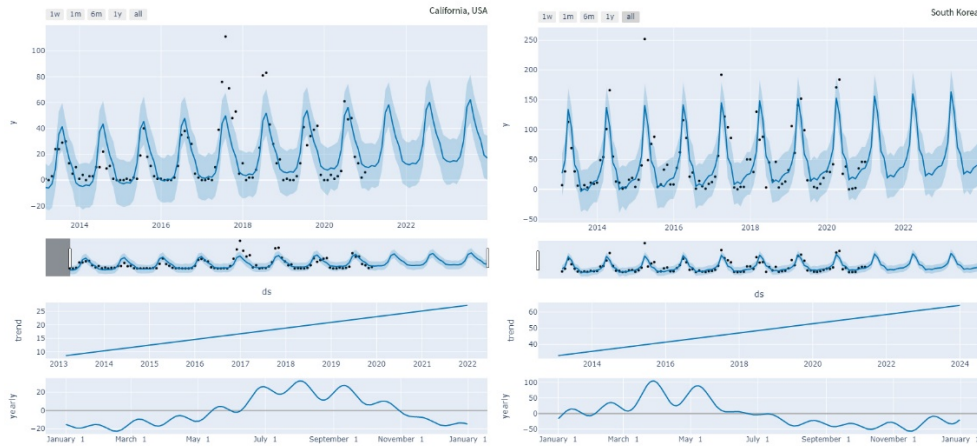


Figure 2: Frequency forecast of wildfire in two regions (California, USA on the left, South Korea on the right)

will have great potential to raise awareness about environmental issues. The main goals of the conventional data visualization are to enhance the understanding of the data, interpret predicted outcome, and suggest possible solutions for improving observed problems. By using interactive multimodal data presentations with more engaging user interactions, users will be able to understand the problems and lead taking actions for more participation to resolve the wildfire. These questions and suggestions have been our inspirations.

Here we propose an interactive multimodal data visualization that explores new ways of visualizing and sonifying the wildfire dataset of two regions (California and South Korea) using contactless user interaction. The two regions were chosen as they have been suffering from the severe wildfire that has occurred in the past seven years. This work aims to explore new aesthetically meaningful visualizations and allow users to learn about the details of wildfires by examining the past and present and predicting the future of the wildfire in the next few years. Since we used temporal data (past, present, and future) and correlation with other related data (e.g., air quality), a proactive spatial portrait using interactive visualization and sonification is necessary to create a more engaging experience. The engagement of users and a proactive mode to understand the topic will enhance the understanding of the potential changes created by actions from the public.

We aimed to create an interactive data visualization that engages the audience and 3D printed sculptures that show a physical representation of past data. The audience can foresee the future of the environmental issues and take any actions by learning the situations with visuals and sound. Furthermore, due to the fear of contact in the pandemic era, the main input device is contactless user interaction using a Leap motion controller. [2] In this paper, we will explore how the past, present, and future wildfire data can be represented in visual, sound, and physical forms. Implementation of our prototype and informal user study from a gallery setup will be discussed.

2 PAST, PRESENT AND FUTURE OF WILDFIRE

We acquired the wildfire dataset of two representative locations from various sources. [3] [4] We then extrapolated this initial model to make a county/city-wise forecasting to obtain statistics by region over the next couple of years. We first selected two regions that have had severe wildfires in the past six years for this research: the (1) State of California and (2) South Korea. The dataset is timestamped, meaning that all its summary statistics could be viewed in a time-series format. This data shows a clear trend and seasonality, making it viable for forecasting. Our evaluation focuses on three factors: the first being the analysis of wildfires in the next couple of years (2021 onwards), the second being how the different attributes of wildfires affect each other and affect the forecast, and the third being the accuracy of our forecast.

We used Prophet [5], an algorithm for forecasting time series data based on an additive model. It is designed for time series with strong seasonal effects and several seasons of historical data. [6] Prophet requires that the data be non-stationary since it uses the trend and seasonality in the data to make a forecast. Thus, no changes were made to the dataset for use in Prophet. After some hyperparameter tuning, the results we received produced the following forecasts. The graphs in figure 2 show an upward trend in the frequency of wildfires in the next couple of years. The dark blue line in both graphs represents the forecasted timeline that the Prophet model created. The light blue area surrounding the dark blue line represents the forecast's saturation capacity and respective saturation floor. It can be considered that the maximum and minimum values towards which the forecast is converging. [7] The yearly seasonality of the observed data shows that California gets most of its wildfires between July and September, while South Korea gets most of its wildfires between March and May. Based on a heuristic visual analysis of the forecasting graphs, we observed that the forecasted plot for frequency captured the trend of the data, and it wasn't affected by the anomalies as such (see: spikes in California wildfire frequencies in 2017) either. Having analyzed the results thus far, it seems clear that the forecasting works well on

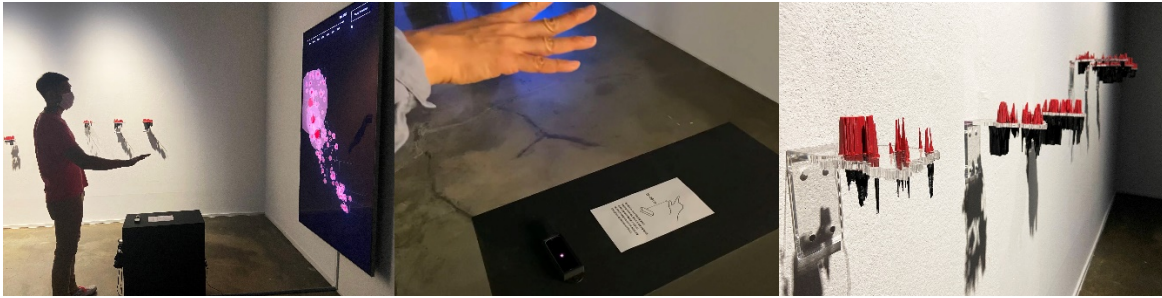


Figure 3: *The Future is Red* on the view at an exhibition in ARKO Art Center, Seoul, S. Korea in 2021. Leap motion controller was used for contactless user interaction. (left and middle). Six 3D printed sculptures were mounted on a wall. (right)

the frequency of fires. However, similar univariate forecasting on the average duration of fires and sum of area burnt doesn't yield similar results, partly because they are summary statistics, and because they have more inconsistencies than the frequency metric. For the purpose of this study, the standard Prophet model has been used to create three-time series models for the three metrics in question. This model has then been optimized for data split across sub-regions in both the State of California and South Korea (county for the former, city for the latter). Thus, the final forecasted data is obtained based on region-specific fire statistics, assuming that each region is independent of the other. Approaches explored towards multivariate time series forecasting (using Vector Autoregression or LSTM Autoencoders) would yield interesting updates to this forecast, and leave some scope for introducing new data related to wildfires (such as population density, air quality index, more detailed and interrelated location data). Based on the results, we exported forecasted data for the data visualization and combined them with the past wildfire data. Therefore, users can observe the past, present, and future wildfire by browsing the detailed information.

3 INTERACTIVE DATA VISUALIZATION, SONIFICATION AND 3D PRINTED SCULPTURES

This project consists of three parts: interactive data visualization, sonification and 3D printed sculptures.

Interactive Data Visualization: As shown in the figure 1, the interactive data visualization shows animated red colored circles that represent each wildfire happened on the geolocation during the given time period (2013–2023) on the map. We chose red color to clearly visualize the fire and give strong impact about the message and awareness about this topic. The time duration of appearance of circles represent the duration of wildfires. The position of the circles represents the location of the wildfire. Opacity and size of the circles represent the acres burned by the wildfires. Overall the users can observe that there are lower number of circles appeared from 2013 to 2016 when there were not many significant wildfires happened. However, toward the present, user can see bigger and opaque, red colored circles appeared on the screen that represent wildfires with longer period and bigger burned area.

Interactive Data Sonification: The digital visualization installation also includes an audio/sound component directly manipulated by

the wildfire data controlled by hand gestures. Users can control both the digital visualization and sonification using hand motions detected by the Leap Motion controller, as figure 3 shows. If no user interacts with the digital visualization, the past wildfire data loops across the screen. However, if a user interrupts and hovers each circle, the animation stops, and users can see a pop-up window on each circle that includes detailed information about the wildfire and triggers sound. Sonification can deliver data storytelling more engagingly. Our target users are the general audience. Thus, we used a simple but effective sound that provides obvious and distinct control parameters rather than subtle details. Frequency, amplitude, duration, and several other parameters of Frequency Modulation (FM) synthesis [8] are the main parameters to create the sound. The Y-axis is mapped to the frequency. The size of the circle is mapped to the amplitude. The duration of the circle animation is mapped to the duration of the sound. Acres burned, casualty, and other impact-related data are mapped to the other parameters of FM synthesis. Therefore, each red dot can create a unique sound, and users can listen to real-time composed harmony or cacophony through hand motions. We used Max [11] for audio synthesis; it goes further by providing multimodal ways of altering and sorting the data in real-time based on the user's intention.

3D printed sculptures: This installation also includes a total of six 3D printed sculptures. Each data sculpture represents previous wildfire data of key years from the whole year ranging from 2013 to 2020. The range of years was selected to represent the recent period of severe wildfire in two regions. The map of the two areas is cut on transparent, recycled plastic plates by laser cutting. Pointy red-colored 3D printed pieces attached above the map represent the wildfire data. Black colored pieces attached underneath the plate represent air quality data of the selected years in the region. Air quality is one of the most correlated data to wildfire. Wildfire impacts the air quality, and air quality also impacts the wildfire. Users can observe the correlation between the two different data in the same region and how they are related to each other.

4 RESULTS AND DISCUSSION

This project was installed at ARKO Art Center, Seoul, South Korea, from September 2021 to December 2021. We tested this prototype at the art gallery and had an informal user study. About twenty participants experienced the installation during the user test and interview. Each participant observed the visualization, listened to the

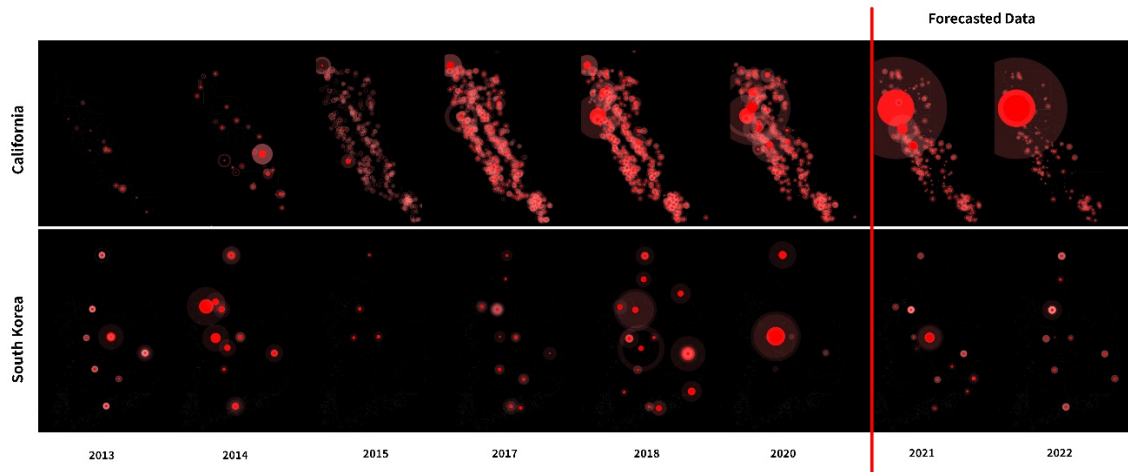


Figure 4: Past, present and future of wildfire in two regions (California, USA on the top, South Korea at the bottom)

sound and interacted with the Leap motion controller. The participants expressed high interest and active engagement in exploring the wildfire data and compared the wildfire of two different regions. They were surprised by the severe wildfire visualization shown in the most recent years (from 2017 to 2020) and the predicted data. (after 2020) Figure 4 shows the past, present, and future of the wildfire in the two regions as a summary. The users also observed the sculptures and looked at the correlation with the air quality. A user mentioned that it was easier to understand the severe impact of wildfire by observing static 3D forms than the 2D animated digital visualization. Users also enjoyed using the Leap Motion controller as they were afraid of touching the surface due to the COVID-19 virus. Although it was hard for some users to adjust the distance between hands and the controller for the first time, most users could browse the data visualization in detailed levels by moving their hands in the air after getting used to the interaction for a while. Overall, the participants understood the relationship between data visualization and data sonification. One user mentioned that high pitch sound with a longer duration occurred when a big circle appeared, creating direct matching results between the visuals and sound. Based on the feedback from the users, we learned that this project could increase public awareness about the current situations of wildfire and share important information about the significant environmental topic by using multimodal interaction. Both analog sculptural element and artificial components were tied well, which represents how humans interact with nature. There were questions and suggestions from the users such as wildfire in other regions, more clear labels on the data sculptures and forecasted wildfire after year 2023, which can be considered as future works.

5 CONCLUSION AND FUTURE WORK

This interactive multimodal data visualization, sonification, and 3D printed sculptures depict the current and predicted situations of the wildfires. Users can observe and explore how the wildfire has been created over time using simple hand motion. The combination of past data representations as 3D printed data sculptures

with correlated air quality data and forecasted data visualization on the screen gives users dynamic multimodal interactive experiences. This project attempts to expand a boundary of multi-sensory experiences for the users in various aesthetic data representations. This data-driven work doesn't require prompt actions for the audience but is more about observations and encouraging them to face the current situations. We believe new ways of visualizing data and storytelling with new media technology will give an impact on the public for their better understanding of climate change.

This is an ongoing project. In keeping with developing a more stable interactive application, a formal user study is necessary to examine how users understand wildfire data visualization and sonification. Wildfire dataset from all other countries/regions will discover any unexpected results and hidden patterns of wildfires in those regions. Regarding the machine learning algorithms and predictive analysis for forecasting the future of wildfire, a more correlated dataset and a more number of attributes will be helpful to understand the situation from diverse angles. Natural disaster data is almost never considered "complete" due to the sheer amount of uncertainty present in nature. Thus, having more related data can make a forecast better capture as much of the uncertainty as possible. Air Quality Index, temperature, and population density of a given region are just some of the factors that could contribute to understanding the frequency of wildfires in the future. Different machine learning modeling and forecasting algorithms can create more interesting results in this domain, especially if trained with more quantitative and qualitative data.

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